Name
Date
Course

# Limits (Part II) Homework

Find each limit using algebraic methods.

① $\lim_{x \to -2} \frac{\sqrt{x+83}-9}{x+2}$

② $\lim_{x \to 25} \frac{x-25}{\sqrt{x}-5}$

③ $\lim_{x \to 0} \frac{\sqrt{x+1} - 1}{x}$

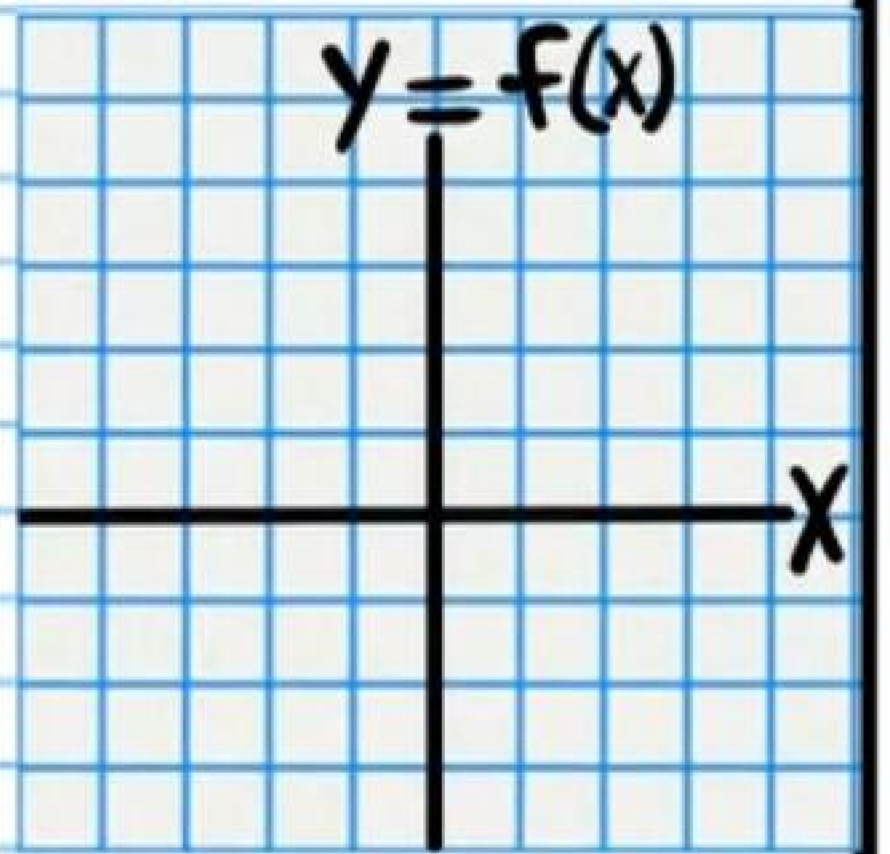
④ $\lim_{x \to 5} \frac{x-5}{\frac{1}{5} - \frac{1}{x}}$

(5) $$\lim_{x\to -3} \frac{\frac{1}{6}+\frac{1}{x-3}}{\frac{1}{x}+\frac{1}{3}}$$

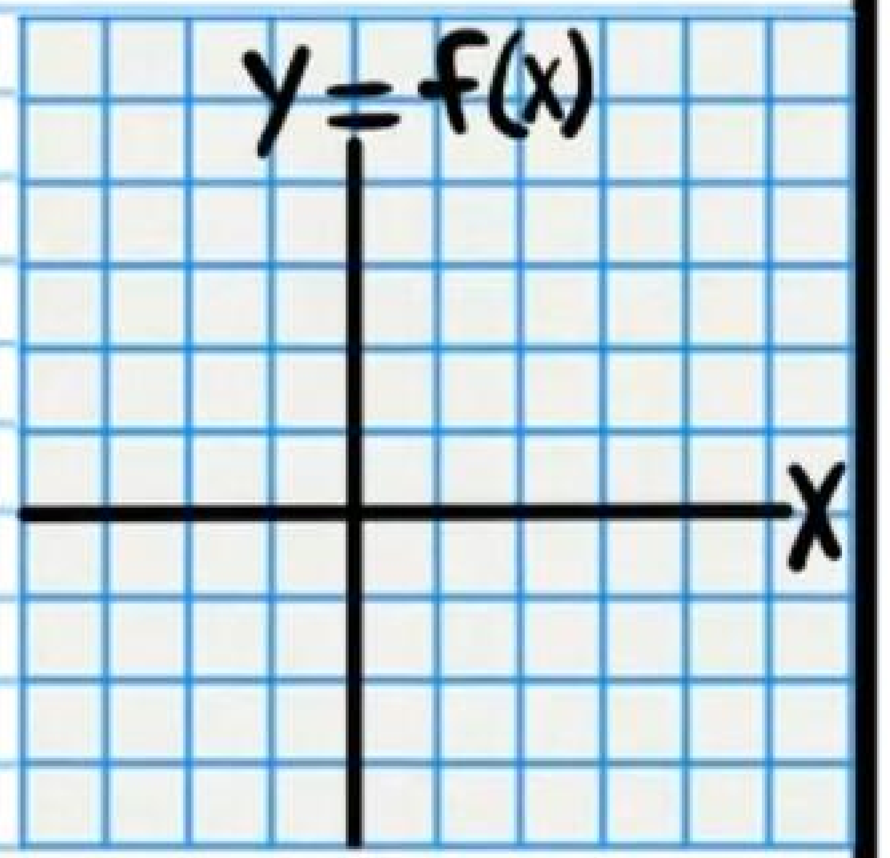
(6) $$\lim_{x\to 0} \frac{2x}{\frac{1}{x+1}-1}$$

Find the value of each limit. You may graph the function if it is helpful.

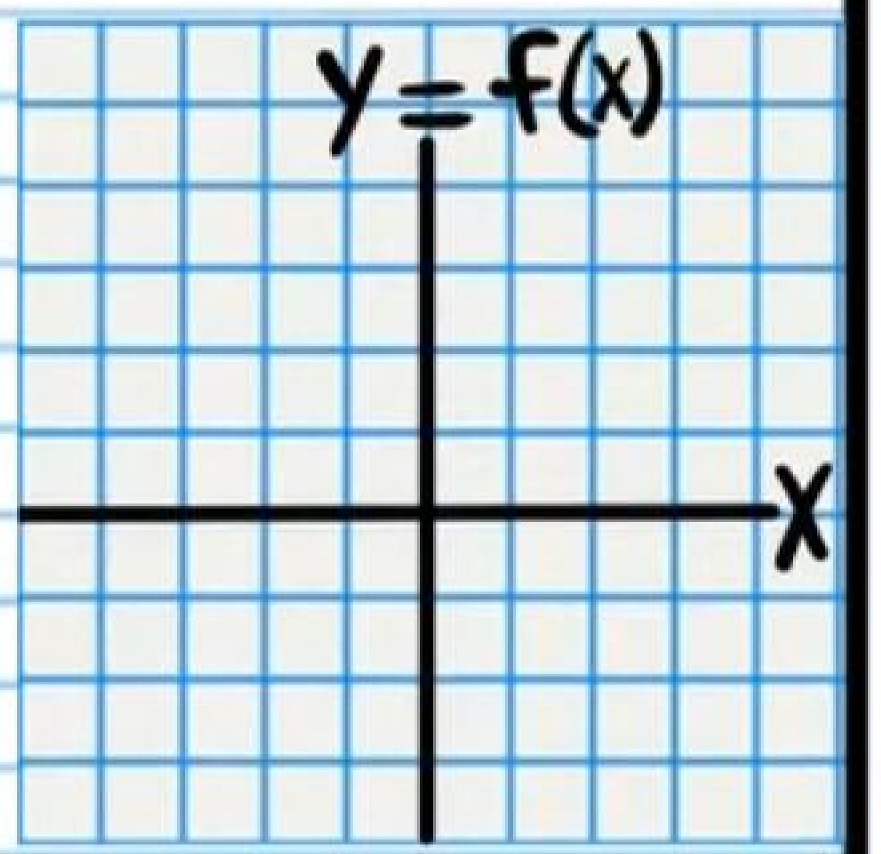
7) $\lim\limits_{x \to -1^+} \frac{1}{x+1}$



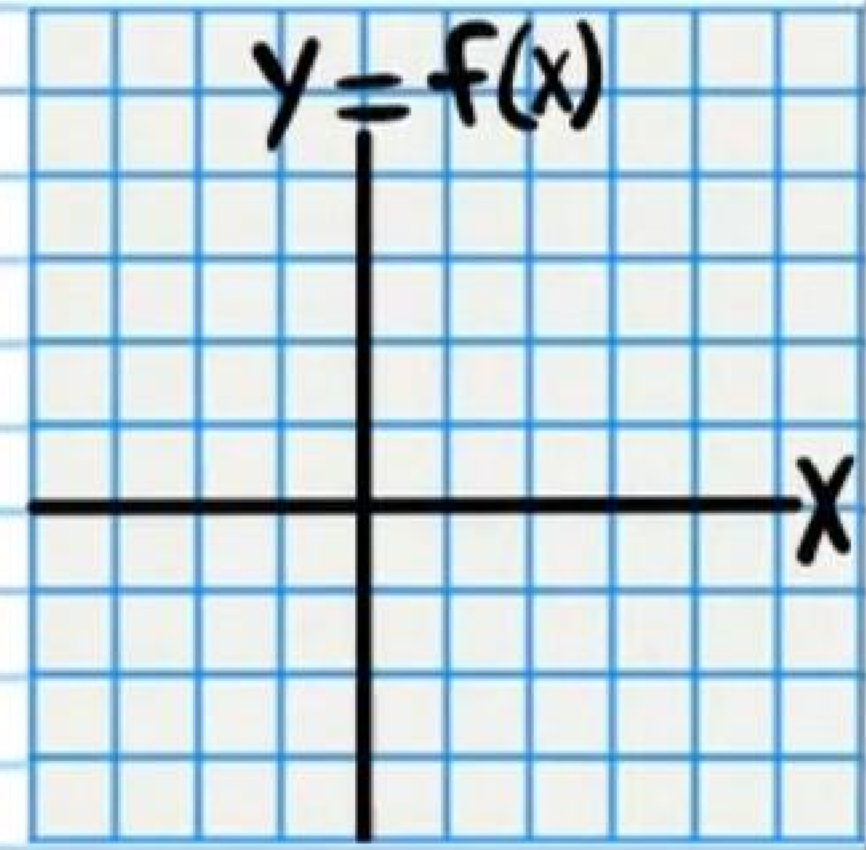

8) $\lim\limits_{x \to 3^-} \frac{1}{x-3}$



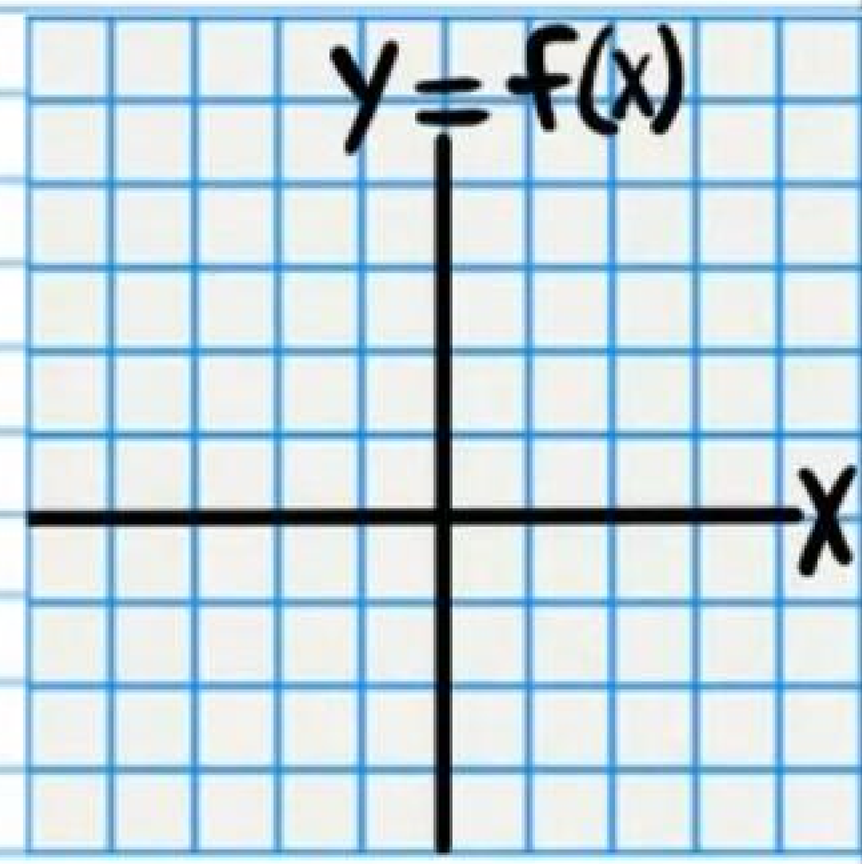

9) $\lim\limits_{x \to -2^-} \frac{-1}{x+2}$



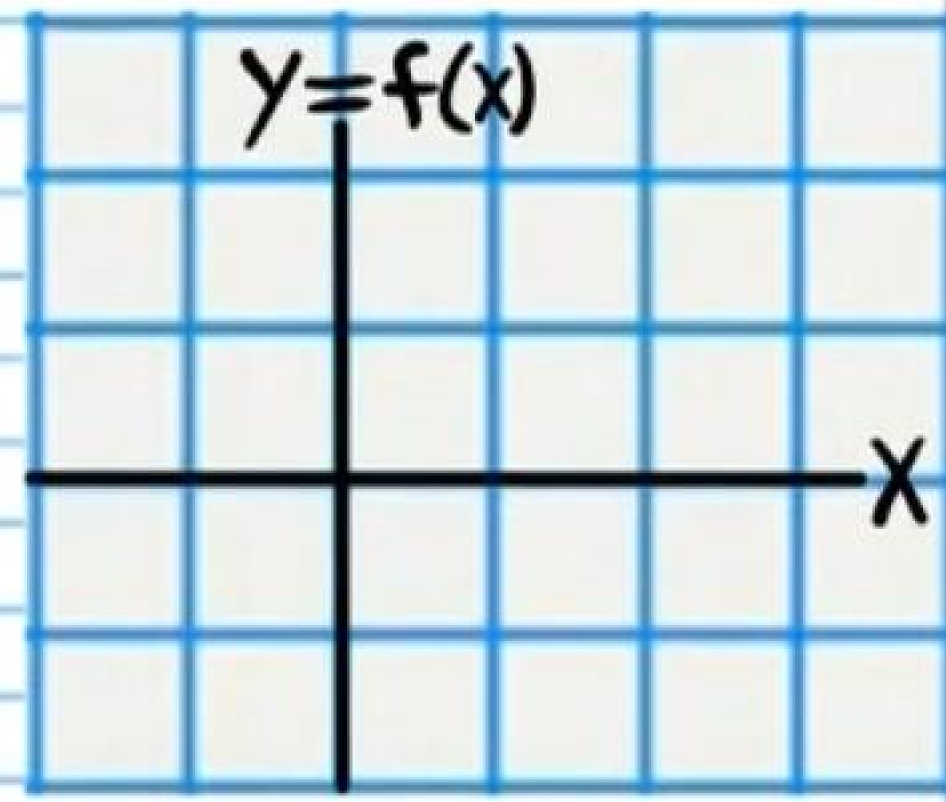

10) $\lim\limits_{x \to 3} \frac{-1}{x-3}$



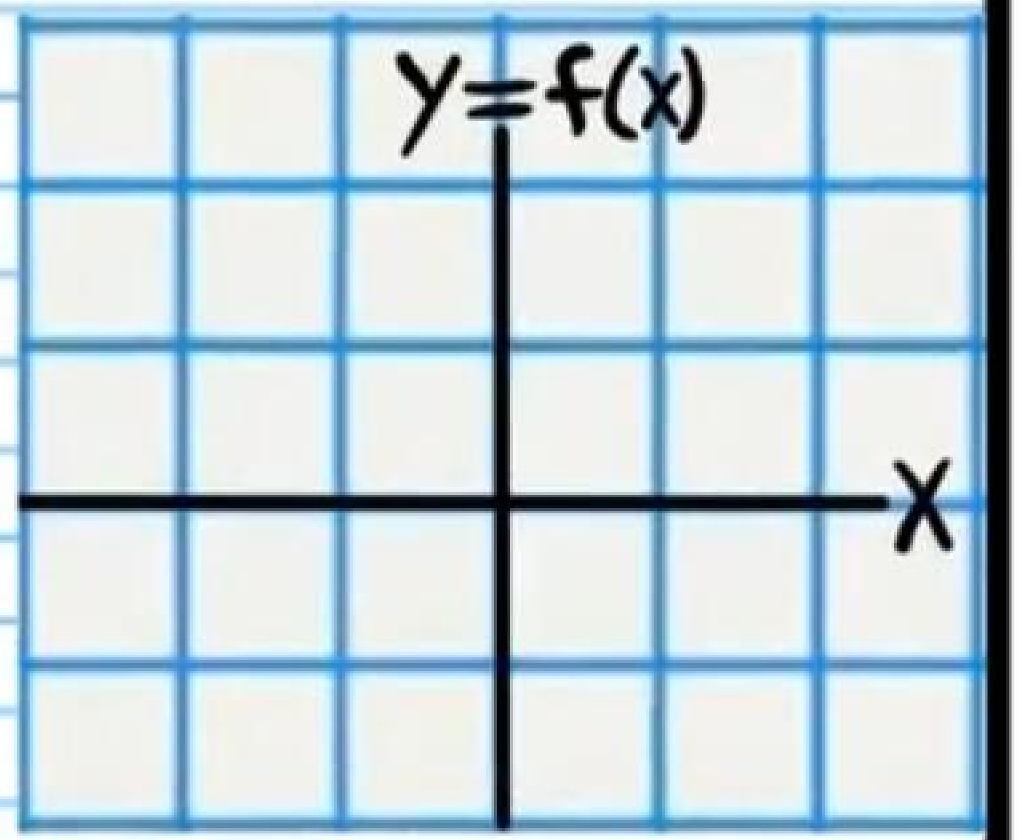

11) $\lim\limits_{x \to -2} \frac{1}{(x+2)^2}$



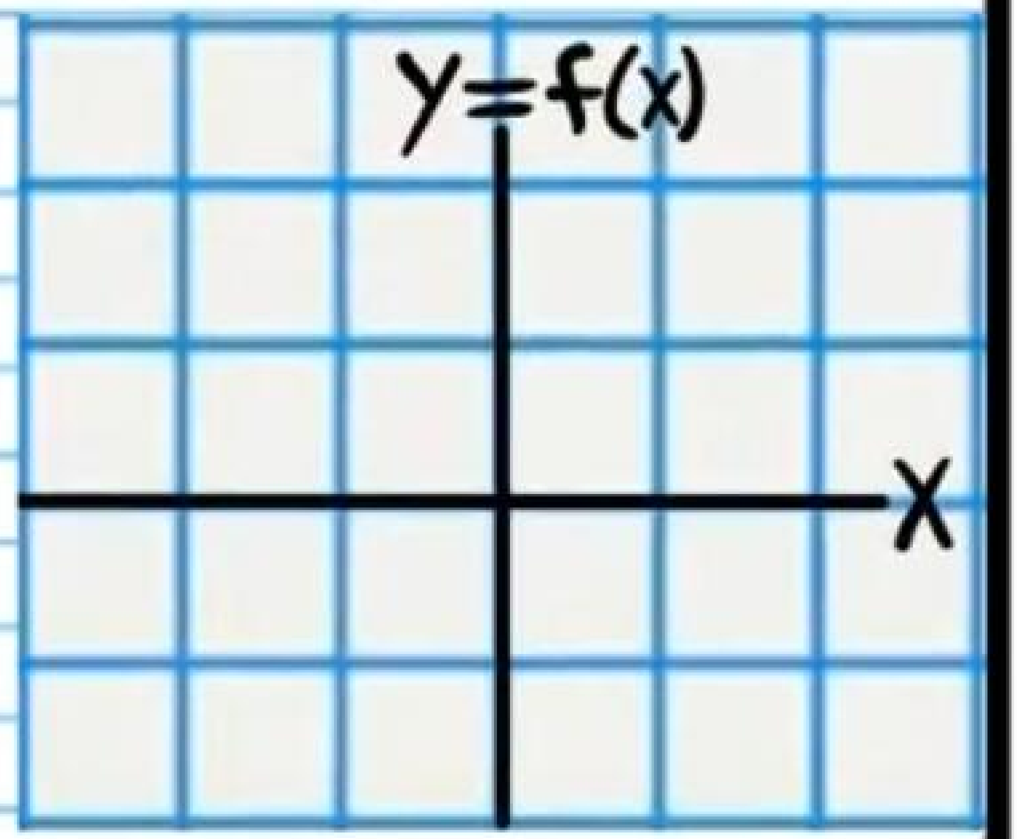

12) $\lim\limits_{x \to \frac{3\pi}{2}^-} \sec x$



(13) $\lim_{x \to -\frac{\pi}{2}^+} \tan x$




(14) $\lim_{x \to 0} \cot x$



# Homework Answers

1. $\boxed{\frac{1}{18}}$
2. $\boxed{10}$
3. $\boxed{\frac{1}{2}}$
4. $\boxed{25}$
5. $\boxed{\frac{1}{4}}$
6. $\boxed{-2}$
7. $\boxed{\infty}$
8. $\boxed{-\infty}$
9. $\boxed{\infty}$
10. $\boxed{\text{DNE}}$
11. $\boxed{\infty}$
12. $\boxed{-\infty}$
13. $\boxed{-\infty}$
14. $\boxed{\text{DNE}}$